AI VOICE COMMUNICATION • SOLUTION PITCH

# VibeLingual

**Beyond Words. Beyond Translation.**

## The problem

Most voice-translation experiences focus primarily on converting spoken words from one language to another. But communication also carries meaning through vocal expression, tone and context. A translation that overlooks these signals can lose part of what a speaker intended to convey.

## Our solution

**VibeLingual** is an AI-powered, emotion-aware Malayalam-to-English voice translation prototype. It brings together speech recognition, acoustic emotion analysis, context-aware translation and English speech generation in one workflow. Its goal is to preserve not only the spoken message, but also useful cues about how it was delivered.

## How it works

| 01 · SPEECH RECOGNITION | 02 · ACOUSTIC EMOTION | 03 · CONTEXTUAL TRANSLATION | 04 · VOICE OUTPUT |
|---|---|---|---|
| Gnani Prisma converts Malayalam speech into text. | Google Gemini analyses the original audio for vocal emotion cues. | Gemini generates an English translation with context and nuance. | Gnani Timbre generates spoken English audio. |

## What makes it different

VibeLingual treats acoustic emotion as a separate signal from the meaning of the transcript. The emotion analysis is based on the original voice recording—not simply on whether the words sound positive or negative. This separation helps avoid assuming that emotionally neutral delivery is sad just because the spoken sentence contains sad content.

## Technology

**Python · Streamlit · Gnani Prisma v2.5 · Google Gemini multimodal AI · Gnani Timbre v2.5**

## Intended value

The prototype explores how voice translation can carry richer conversational context across a language boundary. It is designed as a demonstration of an integrated AI speech pipeline, with opportunities for further evaluation of emotion accuracy, translation quality, latency and accessibility in real-world use.

**Project repository**
github.com/surya-gayathri-s/VibeLingual

**Project tagline**
Beyond Words. Beyond Translation.